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ORIGINAL RESEARCH

Adaptive edge detection of rebar thread head image based on improved Canny operator

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Abstract
The quality of the rebar thread head is crucial for the connection between the rebars, thereby affecting the public safety of the building. The edge extraction of the thread image is the most important step in obtaining the accurate size parameters, which helps to complete online inspection of quality. Here, an edge detection method based on improved Canny operator is proposed to solve the problems on sensitivity to noise, existence of false edges, and lack of self-adaptability. Firstly, an improved adaptive median filtering algorithm is used to denoise the image. Then replace Sobel with Scharr to enhance the gap between pixel values, add 45° and 135° directional templates to prevent edge information loss as well. Besides, introduce a scale factor to improve the interpolation method for non-maximum suppression and reduce false edges. Finally, use Ostu operator to achieve adaptive extraction of high and low thresholds to complete adaptive edge detection. The qualitative analysis and quantitative results show that the improved algorithm can not only have a good visual effect, have good robustness and feasibility, but also be faster than other algorithms, which provide a basic guarantee for real-time online quality detection of the thread head.

1 | INTRODUCTION

The continuous innovation and progress of rebar mechanical connection technology have been widely applied in engineering fields such as housing construction, municipal engineering, rail transit, bridges and tunnels, hydraulic structures, nuclear power plants and so on [1]. As an important component of mechanical connection, the quality of rebar processing is crucial to the quality and safety of the project. The image of the thread head used for rebar connection is shown in Figure 1. With the improvement of rolling automation and the continuous acceleration of production rhythm, the traditional method for thread quality sampling inspection based on manual vision has high work intensity, low efficiency, low accuracy, and the risk of missing inspection, which has been difficult to meet the needs of modern production. With the development of digital measurement technology, non-contact detection technology based on machine vision has been widely used in the industrial field. It has become a trend to use image processing technology as the core method for online detection of the quality of rebar thread head. An important feature of an image is the edge. Obtaining edge features is an important way to identify size information such as the major diameter, pitch diameter, and minor diameter of thread heads. The quality of edge detection directly affects the final result of size measurement. Therefore, an effective edge detection method is crucial for obtaining the above dimensional parameters [2].

In an image, the edge of an object often refers to an area where the grayscale value of the image suddenly changes. The edge has two attributes: direction and amplitude, which can usually be detected by first or second derivative detection. The first derivative takes the maximum value as the position of the corresponding edge, while the second derivative takes the zero-crossing point as the position of the corresponding edge. Domestic and foreign researchers have done a lot of work on edge detection tasks, which can be divided into traditional edge detection algorithms and modern ones. There are many traditional edge detection algorithms, such as Sobel operator, Roberts operator, Laplacian operator, LOG operator, Prewitt operator, Kirsch operator and so on [3]. The basic principle of the above algorithms are to calculate the gradient of the local area window, which have high real-time performance and low implementation cost. However, it is highly sensitive to noise in the detection image and has low signal-to-noise ratio. With the rapid development of machine learning technology, some methods based on statistical machine learning [4–6] have been introduced into image contour and edge detection. These methods usually use convolution or similar convolution to perform grayscale analysis on digital images to determine the edges of the image. However, these methods still cannot effectively extract semantic information affected by complex image content [7]. Convolutional Neural Network (CNN) has made significant progress in the field of computer vision, such as image recognition [8–9], object detection [10], semantic segmentation [11] etc. CNN has the ability to automatically extract high-quality image semantic information, and a series of edge detectors based on CNN have been proposed in image edge detection [12–13]. The above modern edge detection algorithms have high robustness and high edge positioning accuracy, which are more accurate than traditional edge detection algorithms. However, it requires a large amount of data for training, which results in high detection costs and are not suitable for online detection [14]. John Canny proposed three criteria for evaluating edge extraction: signal-to-noise ratio criteria, positioning accuracy criteria and unilateral response criteria. Based on these three criteria, he proposed the Canny operator [15]. The Canny operator has better accuracy than the traditional methods, moreover, there are not as many requirements for the amount of image data. However, there are still issues such as unclear and incomplete edge contours, sensitivity to noise, and low signal-to-noise ratio. In addition, it is necessary to manually select high and low thresholds during the detection process, resulting in significant randomness in selecting relevant parameters, which leads to reduced accuracy in detection.

Canny edge detection is divided into four steps, including filtering and denoising, gradient calculation, non-maximum suppression, and dual threshold edge point extraction. Many scholars at home and abroad had made various improvements to the structure and methods of the Canny operator to improve the accuracy and universality of edge detection. In terms of filtering and denoising, multiple filters were proposed to replace Gaussian filters to smooth and denoise images to improve the signal-to-noise ratio of images, such as local linear kernel smoothing filters, statistical filters, wavelet transforms, morphological filters, adaptive median filters, fast guidance filters and so on [16–24], which had effectively suppressed noise and achieved good results. In addition, a dual filter combining adaptive fuzzy median filtering and bilateral filtering was proposed to eliminate salt and pepper noise in images [25]. Experimental

FIGURE 1 Rebar head in kind.

results showed that the dual filter had effectively improved the image signal-to-noise ratio and ensured image edges. In gradient calculation, in order to prevent edge information loss and further reduce noise interference, many improved algorithms focus on using the 3×3 finite difference method instead of the 2×2 finite difference method in the traditional Canny operator, and calculating pixel gradients in the 45°, 135° diagonal directions or more by adding gradient templates [26–29], which have improved edge noise immunity to a certain extent, but the positioning accuracy reduced. Non maximum suppression is mainly used to locate the image and reduce the number of false edges in the image. In the process of non- maximum suppression, the main method to obtain edge points is to obtain local maximum values through interpolation and comparison. However, the selection of interpolation points by traditional Canny operators is very arbitrary, resulting in inaccurate edge detection. Therefore, the improvement was mainly achieved by improving the interpolation algorithm to achieve the maximum gradient [30–32]. Proper selection of the high and low threshold values in the Canny operator would achieve better edge detection results for the image. Generally speaking, the traditional Canny operator usually manually selects high and low thresholds, which results in the unobvious feature of edge points obtained. To overcome this problem, many improved algorithms for automatically and semi-automatically determining high and low thresholds have been proposed, such as one-dimensional entropy method, improved histogram, Newton iterative method, and 1D/2D Otsu method [33–37]. These methods were based on edge length, gradient size, gradient direction and average distance between edge bifurcations. As a result, a lot of edge pixel recognition and connection methods with certain applicability were proposed in different edge detection environments. In summary, the improvement methods of traditional Canny edge detection are very rich, and are still developing and improving.

The Canny algorithm should be improved in specific scenes in the existing research, and the detection effect is affected by specific application fields. The purpose of edge detection to rebar thread head image is to accurately and quickly extract thread size parameters online, and complete the quality detection. Therefore, edge detection methods are required to be fast and accurate. In the process of image acquisition, due to the influence of external factors (on-site environment, light intensity) and other objective factors, there are problems such as high noise and residual coolant on the surface of the image, making it easy to appear some isolated pixel points or blocks with strong visual effects on the image, thereby disturbing the observability of the image, affecting the edge feature extraction of the thread image, and seriously affecting the acquisition of subsequent parameters to a certain extent. In addition, Canny algorithm is usually used to process images without considering the impact of parameter configuration and specific implementation methods at each step. However, in different application scenarios, the optimal parameters are different. Therefore, it is necessary to further study the application of Canny edge detection algorithm in the detection of the rebar thread image.

In order to better eliminate noise and protect the detailed features of the image edges from being blurred, and effectively and quickly extract the edge information in rebar thread image, the application and improvement of Canny edge detection algorithm is discussed. Section 2 introduces the principle and steps of traditional Canny algorithm. In Section 3, an improved Canny operator is proposed, along with the specific implementation of each step and the optimal selection of methods. In Section 4, through experiments and simulations, the effectiveness of the improved Canny algorithm is verified. Section 5 concludes the article.

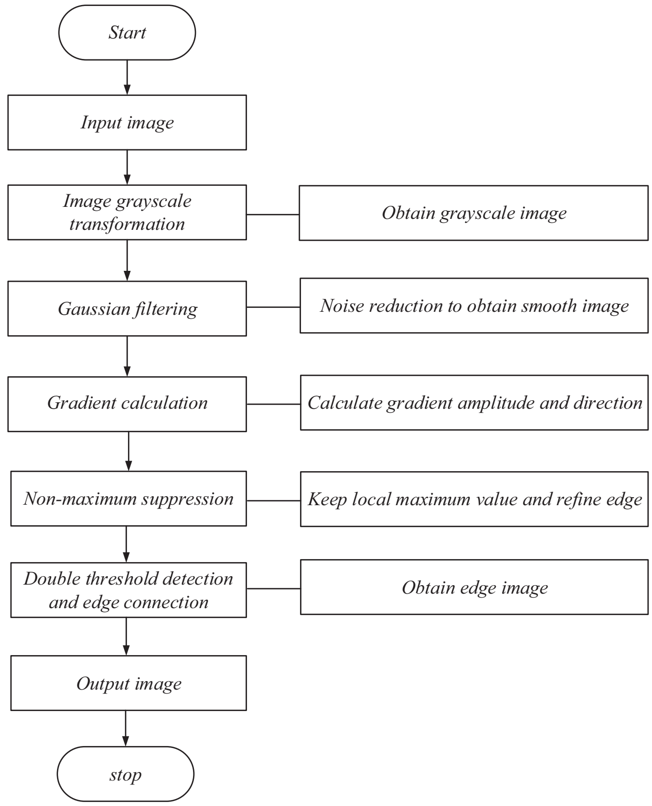

FIGURE 2 Flow chart of classical Canny operator.

2 | TRADITIONAL CANNY OPERATOR

The Canny operator is a multi-level detection algorithm proposed by John F. Canny. Generally, the image processed is a grayscale image. If the input image is a colour image, it needs to undergo image grayscale conversion to become a grayscale image for subsequent image processing. The output image is an edge detection gradient binary image. The basic process is shown in Figure 2.

2.1 | Gaussian filter to smooth image

The Canny operator first uses a Gaussian filter to remove image noise before processing, and uses the first derivative of a 2D Gaussian function to smooth the image [38]. The 2D Gaussian function is shown in Equation (1).

$$G\ (x,y) = \frac{1}{2\pi\sigma^2}\ \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{1}$$

where the gradient vector is as follows:

$$\nabla G\ (x,y) = \begin{bmatrix} \dfrac{\partial G}{\partial x} \\ \dfrac{\partial G}{\partial y} \end{bmatrix} \tag{2}$$

In order to improve the operational speed of the Canny operator, the 2D convolution templates of $\nabla G(x,y)$ is decomposed into two one-dimensional filters, which are shown in Equations (3) and (4).

$$\frac{\partial G\ (x,y)}{\partial x} = kx \cdot \exp\left(-\frac{x^2}{2\sigma^2}\right) \exp\left(-\frac{y^2}{2\sigma^2}\right) \tag{3}$$

$$\frac{\partial G\ (x,y)}{\partial y} = ky \cdot \exp\left(-\frac{y^2}{2\sigma^2}\right) \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{4}$$

Convolve these two templates with the image $f(x,y)$ to obtain E_x and E_y.

$$E_x = \frac{\partial G\ (x,y)}{\partial x} \times f\ (x,y) \tag{5}$$

$$E_y = \frac{\partial G\ (x,y)}{\partial y} \times f\ (x,y) \tag{6}$$

where $A\ (i,j) = \sqrt{E_x^2(i,j) + E_y^2(i,j)}$, $\alpha\ (i,j) = \arctan\frac{E_y(x,y)}{E_x(x,y)}$, $A(i,j)$ reflects the edge intensity at the (i, j) point on the image, $\alpha\ (i, j)$ is the direction of the edge. σ is a parameter which affects the quality of the Gaussian filter. if σ is small, the detection accuracy is relatively ideal, but the filtering effect is general, and the position of edge points in the actual detection results will have a large deviation from the actual position; if σ is large, it is significant to smooth the noise, but

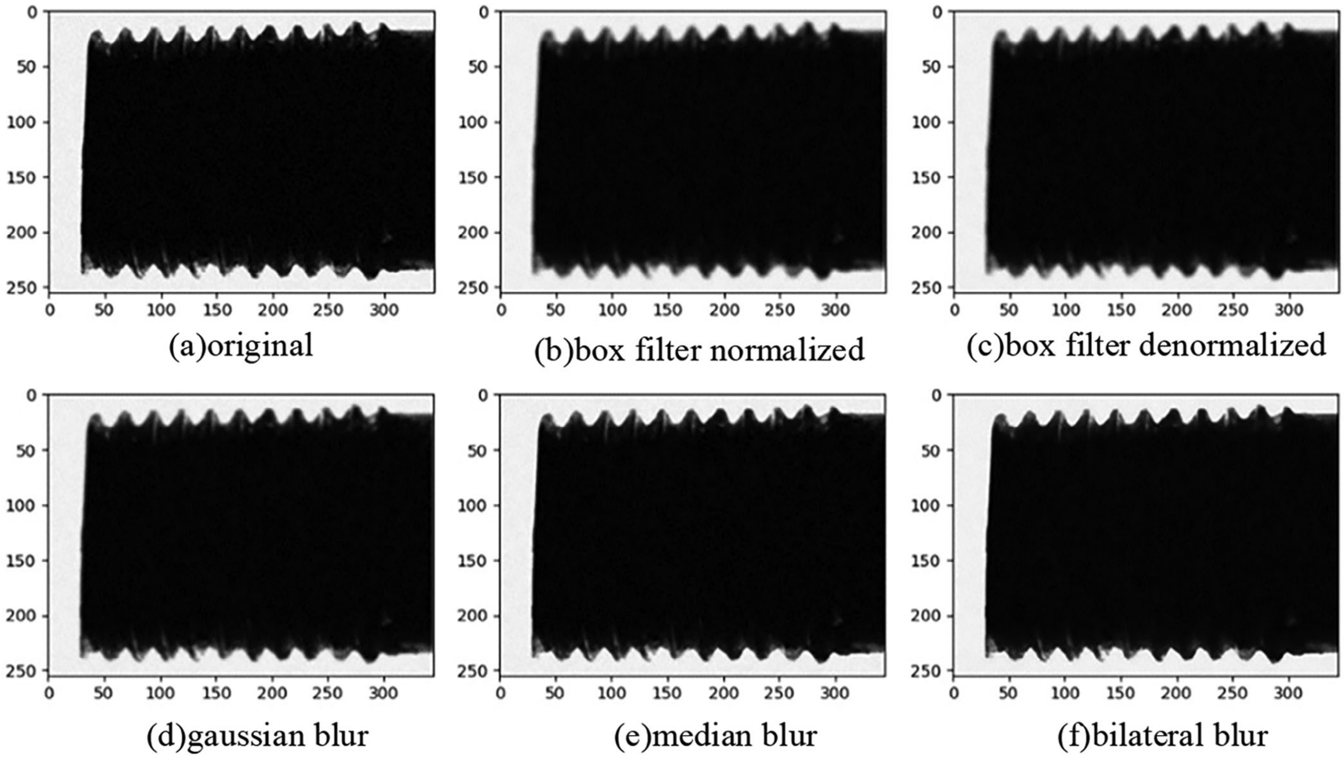

FIGURE 3 The processing effect of classical filter operators on thread head images. (a) original; (b) box filter normalized; (c) box filter denormalized; (d) gaussian blur; (e) median blur; (f) bilateral blur.

can lead to blurred edges of the image, which will result in low positioning accuracy for edges. Besides, the value of σ needs to be manually set, which has significant limitations.

2.2 | Gradient calculation

The traditional Canny operator uses finite differences with 2 × 2 neighbourhoods in both horizontal and vertical directions to obtain the gradient of data array $M(x,y)$. Partial derivative in both directions at point (x, y) defined as $N_x(x,y)$ and $N_y(x,y)$ are expressed in Equation (7) and (8), respectively.

$$N_x\ (x,y) = \frac{[M\ (x,y+1) - M\ (x,y)]}{2} + \frac{[M\ (x+1,y+1) - M\ (x+1,y)]}{2} \tag{7}$$

$$N_y\ (x,y) = \frac{[M\ (x,y+1) - M\ (x+1,y)]}{2} + \frac{[M\ (x,y+1) - M\ (x+1,y+1)]}{2} \tag{8}$$

The calculation of derivatives is very sensitive to noise. It is easy to make Mis-inspection if the edge detection is performed on an image only from horizontal and vertical directions. The gradient amplitude and gradient direction of a pixel are calculated with the transformation formula from rectangular coordinates to polar coordinates, and the gradient amplitude and direction are calculated with the second order norm in Equations (9) and (10), respectively.

$$Q\ (x,y) = \sqrt{N_x(x,y)^2 + N_y(x,y)^2} \tag{9}$$

$$P\ (x,y) = \arctan\left(\frac{N_y\ (x,y)}{N_x\ (x,y)}\right) \tag{10}$$

If the gradient amplitude $Q\ (x, y)$ of a pixel point (x, y) is greater than or equal to the gradient amplitude of two adjacent pixel points along the gradient direction $P\ (x, y)$, it can be determined that the point is a possible edge point.

2.3 | Non maximum suppression

In order to obtain more accurate edge points of the rebar thread head image, it is necessary to subdivide them to find the edge points with the most prominent amplitude fluctuations. The processing flow for non-maximum suppression is as follows: first, find the non-zero point of the gradient intensity, and find two adjacent points along the gradient direction of the point. If the amplitude of these two adjacent points is greater than their starting point, it can be determined that the point is not on the edge, otherwise, it can be used as a candidate edge point.

TABLE 1 Comparison of filtering effects.

		Box filter normalized	Box filter denormalized	Gaussian blur	Median blur	Bilateral filter
MSE	Value	152.786	152.786	73.22	10.144	13.463
	Time (s)	0.247	0.243	0.243	0.271	0.292
PSNR	Value	26.29	26.29	29.485	38.69	36.839
	Time (s)	0.001	0.001	0.002	0.002	0.002
SSIM	Value	0.894	0.894	0.947	0.965	0.923
	Time (s)	0.016	0.019	0.015	0.014	0.012

2.4 | Dual threshold detection and edge connection

The algorithm of dual threshold uses high and low thresholds for segmentation of images that have undergone non maximum suppression. If the edge strength value of a point is larger than the high threshold value, the point is marked as an edge point; If it is smaller than the low threshold value, it is recorded as a non-edge point; If it is between a high threshold and a low threshold, it is necessary to determine whether the pixel point is connected to the previously obtained edge point; If connected, the pixel point is marked as an edge point, and vice versa. In this way, by continuously marking edge points, unclosed edge points can be connected into edge contours, thereby obtaining the edges of the image.

3 | IMPROVEMENT OF TRADITIONAL CANNY OPERATOR

3.1 | Improved adaptive weighted median filtering design

A Gaussian filter is used to smooth rebar thread head images in the traditional Canny operator, which requires manual selection of parameters during the processing. Therefore, it is often difficult to obtain an ideal noise removal thread head image due to improper parameter selection. The purpose of smooth filtering for rebar thread head images is to eliminate noise and improve the signal-to-noise ratio. However, in fact, during the process of filtering, some edge information will be processed, resulting in a decrease in the number of edges. In order to effectively perform adaptive image filtering and achieve rapid noise reduction of images, several classic filter operators are combined to process grey scale images of thread head, and study suitable filtering algorithms for rebar thread head. The results are shown in Figure 3.

There are many ways to evaluate the quality of an image. Currently, the most commonly used methods are mean square error (MSE), peak signal to noise ratio (PSNR), and structural similarity (SSIM) [39, 40].

MSE is the mean value of the sum of squares of the differences between the predicted value $f(x)$ and the target value y. The smaller the mean square error value, the closer the processed rebar thread image is to the original image. Assume there are two monochrome images I and K, whose sizes are both $m \times n$. The definition for MSE can be shown in Equation (11).

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \left[I(i,j) - K(i,j) \right]^2 \tag{11}$$

PSNR is the peak signal to noise ratio. The higher the peak signal to noise ratio, the greater the amount of information in the processed rebar thread head image, the less surface noise, and the clearer the thread head image. The PSNR of two images is defined as follows:

$$PSNR = 20\log_{10}\left(\frac{MAX_i}{\sqrt{MSE}}\right) \tag{12}$$

SSIM is the structural similarity, whose principle is to evaluate the similarity of local area content between images. It can measure the differences between enhanced images and real images, thereby guiding the learning process. Assuming that the two input images are X and Y, the definition for SSIM is expressed as follows:

$$SSIM(X,Y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\delta_{xy} + C_2\right)}{\left(\mu_x^2\mu_y^2 + C_1\right)\left(\delta_x^2\delta_y^2 + C_2\right)} \tag{13}$$

where μ_x and μ_y represents the average of X and Y, respectively; δ_x^2 and δ_y^2 represents the variance of X and Y, respectively; δ_{xy}represents the covariance of X and Y; $C_1$and C_2 are constants to prevent the value of the denominator is 0. The larger the SSIM value, the smaller the difference between the output image and the undistorted image, which means the better the image quality.

In order to illustrate the effect of the above filtering algorithms on noise processing, the experimental comparison results of the MSE, PSNR, and SSIM obtained from the original rebar thread head image through mean filtering, square box filtering, Gaussian filtering, median filtering, and bilateral filtering are shown in Table 1. From the table, it can be found that the average values of the images processed by the above five algorithms all get improved. From the results of PSNR, the median filter has the largest value, representing the smallest difference between the processed image and the original image. As for

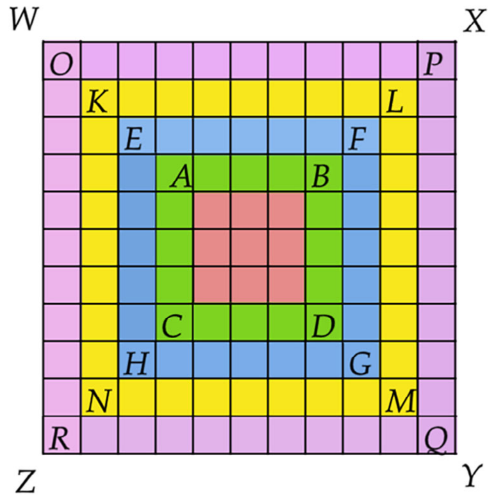

FIGURE 4 Enlargement rule of adaptive median filtering window.

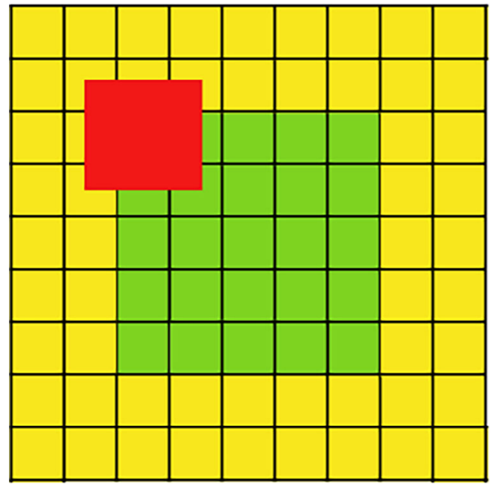

FIGURE 5 Schematic diagram of expanded image.

TABLE 2 Number of repeated sequencing points under different window iterations.

Window size	Number of repeated sorting points
3×3–5×5	9
3×3–7×7	34
3×3–9×9	83
3×3–11×11	164

SSIM, the value of median filtering is the largest as well. Synthetically, median filtering has the best filtering effect. In terms of speed, median filtering processing is also relatively fast. Due to the fact that the edge quality of the original image of the rebar thread head affects the subsequent contour extraction, median filtering can complete the filtering operation while protecting the signal edge. Therefore, median filtering is selected as the filtering algorithm for the image of rebar thread head.

The basic principle of median filtering is to sort the points within a certain window range in a digital image (the window size is usually an odd number), and then take the intermediate value in the sorting sequence as the grey value of the pixel points in the centre of the window. The mathematical expression of median filtering is shown in Equation (14).

$$I\ (x, y) = \ median \left\{ f\ (k)\ , k \in W \right\} \tag{14}$$

where $I(x, y)$ represents the processed image, k represents the number of pixels in the window, W represents the template, $f(k)$ represents the sorted grayscale value sequence, and median represents the median value of the set.

For traditional median filtering, when the probability of black and white noise spots in the image is greater than 20%, the filtering effect will decline sharply. This is because traditional median filters do not "distinguish" the noise spots within the window. When the proportion of invalid noise points within the window exceeds 50%, the median points will be taken as noise points, making the filtering meaningless. After determining the initial filtering window size, adaptive median filtering will establish a two-level loop to determine whether the obtained median point is a noise point. At the same time, the window will be adaptively expanded according to the size of "$W + 2$" to continue filtering, effectively avoiding the filtering failure problem in median filtering. Noise reduction filtering is performed based on the extreme value points of the adaptive window as the judgment basis for noise points, which has the following three defects:

1. The adaptive window takes the point of the image as the centre point, and uses the template of "$N \times N$" to get gradually filter, ignoring the pixel points adjacent to the outer edge of the image, affecting the overall noise reduction effect of the image.
2. During the "$W + 2$" expansion iteration of the window, a portion of pixel points will repeatedly participate in the operation, as shown in Figure 4. When the filtering window is expanded from square ABCD (3×3) to EFGH (5×5), the $3 \times 3 = 9$ pixels contained in the former will be repeatedly involved in the sorting calculation. Similarly, if the window expands to KLMN (9×9), there will be 9 points to be sorted three times, 25 points to be sorted two times, and 49 points to be sorted one time, that is, 83 points will be sorted repeatedly in one operation, which greatly reduces the efficiency of the algorithm. When the lower limit of window size is 3×3 and the upper limit is 11×11, the number of repeated sorting points in different windows are as shown in Table 2.
3. For windows that are judged to contain noise points, the median grey value of the pixel points in the region where they are located is taken as the filtering result. The size of the median I_{med} has a significant impact on the filtering output.

Here, an improved adaptive median filtering noise reduction method is put forward. This method expands the image size before filtering, and quickly determines the window size based on distinguishing between noise and effective pixel points. If there are no noise points in the window area, the original pixel

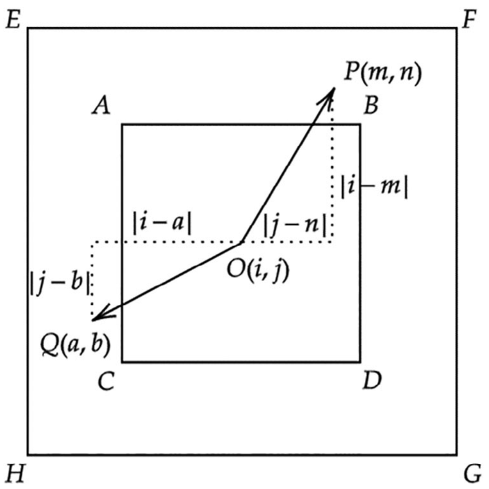

FIGURE 6 Determination of window size.

points on the image are used for output. To the contrary, the noise points 0 and 255 are removed during sorting, and only the effective pixel points in the window are sorted. The weighted value is used to replace the median noise point, which improves the filtering effect. Specific improvements are as follows:

1. Image boundary extension: To avoid filter templates exceeding the boundaries of the image during convolution calculations, traditional filtering algorithms usually ignore noise reduction filtering on image boundary pixels, resulting in blurred image boundaries. This algorithm expands the image size before filtering. Assume that the size of the original image $f(x, y)$ is $M \times N$, take the image boundary as the axis of symmetry, and copy several rows or columns adjacent to the boundary as the expanded pixel points of the image. After boundary expansion, the size of the image becomes as $(M + 2N_{max}) \times (N + 2N_{max})$, and the position of a pixel on the original image changes from $w_{i,j}$ to $w_{i+N_{max}, j+N_{max}}$, the gray value corresponding to the pixel point does not change, and the expanded image is shown in Figure 5.

1. Quickly determine the optimal window size: This algorithm quickly determines the optimal filter window size through the distance between the effective pixel point and the centre point. As shown in Figure 6, the window size is determined to be 5×5 by directly measuring the size of the coordinate distance r between the centre point O and the effective pixel point P, effectively reducing the number of sorting times, and the optimization effect is particularly evident when the window is large. When comparing the values of r, both the abscissa difference $r = |i - m|$ and the ordinate difference $s = |j - n|$ should be compared, make $\{c = \max(r, s)\} \in EFGH$. Similarly, the difference of the horizontal and vertical coordinates between the effective pixel point Q and the centre point O should also be calculated to ensure that the maximum value is within the window area. Finally, the window size $L \geq \max_k(r, s)$, where k is the number of effective pixel points within the window. When the window is large, this improvement significantly reduces the number of algorithms, as shown in Table 3.

TABLE 3 Complexity comparison of algorithms.

Window size	Classical adaptive median filter	Improved adaptive median filter
3×3	36	≤36
5×5	336	≤300
7×7	1512	≤1176
9×9	4752	≤3240
11×11	12012	≤7260

1. Operation on valid pixel points: The improved algorithm does not perform filtering for valid signal points, which saves a lot of unnecessary running processes compared to the traditional algorithm of storing and sorting before discrimination. Take the window size as a window containing more effective pixel points will make the value of the replaced pixel points more reasonable. When sorting, remove noise points, and take the weighted median value in the window composed of effective pixel points, which improves the filtering effect. The specific steps are as follows.

Step 1 Take the point $O(i, j)$ on the image as the centre point of the window, sort the grayscale values of the effective pixel points within the window to find the median point, I_{med} and average point I_{mean};

Step 2 Take the sorted median point as the centre point of the current adaptive window, multiply the effective pixel point values in the neighbourhood of the window by their normalized weighting coefficients, obtain the sum, and take this value as the filtered output value of the current noisy window, which can be shown in Equation (15).

$$\hat{f}(i, j) = \sum_{(i-r, j-r) \in w_r} w(i - r, j - r) * f(i - r, j - r) \tag{15}$$

where $w_r (r \in (-N_{max}, N_{max}), i = 1, \ldots, M, j = 1, \ldots, N)$ represents an adaptive filtering window, $\hat{f}(i, j)$ is the filtered pixel value, and the weighting coefficient $w(i, j)$ can be described as Equation (16).

$$w(i, j) = \frac{\frac{1}{1+\max\{S_{mean}, (I(i-r, j-r) - I_{med})^2\}}}{\sum_{(i-r, j-r) \in w_r} \frac{1}{1+\max\{S_{mean}, (I(i-r, j-r) - I_{med})^2\}}} \tag{16}$$

where S_{mean} is the average value of the square of the difference between the effective pixel values and their mean values in the filtering window, as shown in Equation (17).

$$S_{mean} = \frac{\sum_{(i-r, j-r) \in W_r} (I(i - r, j - r) - I_{mean})^2}{N_r} \tag{17}$$

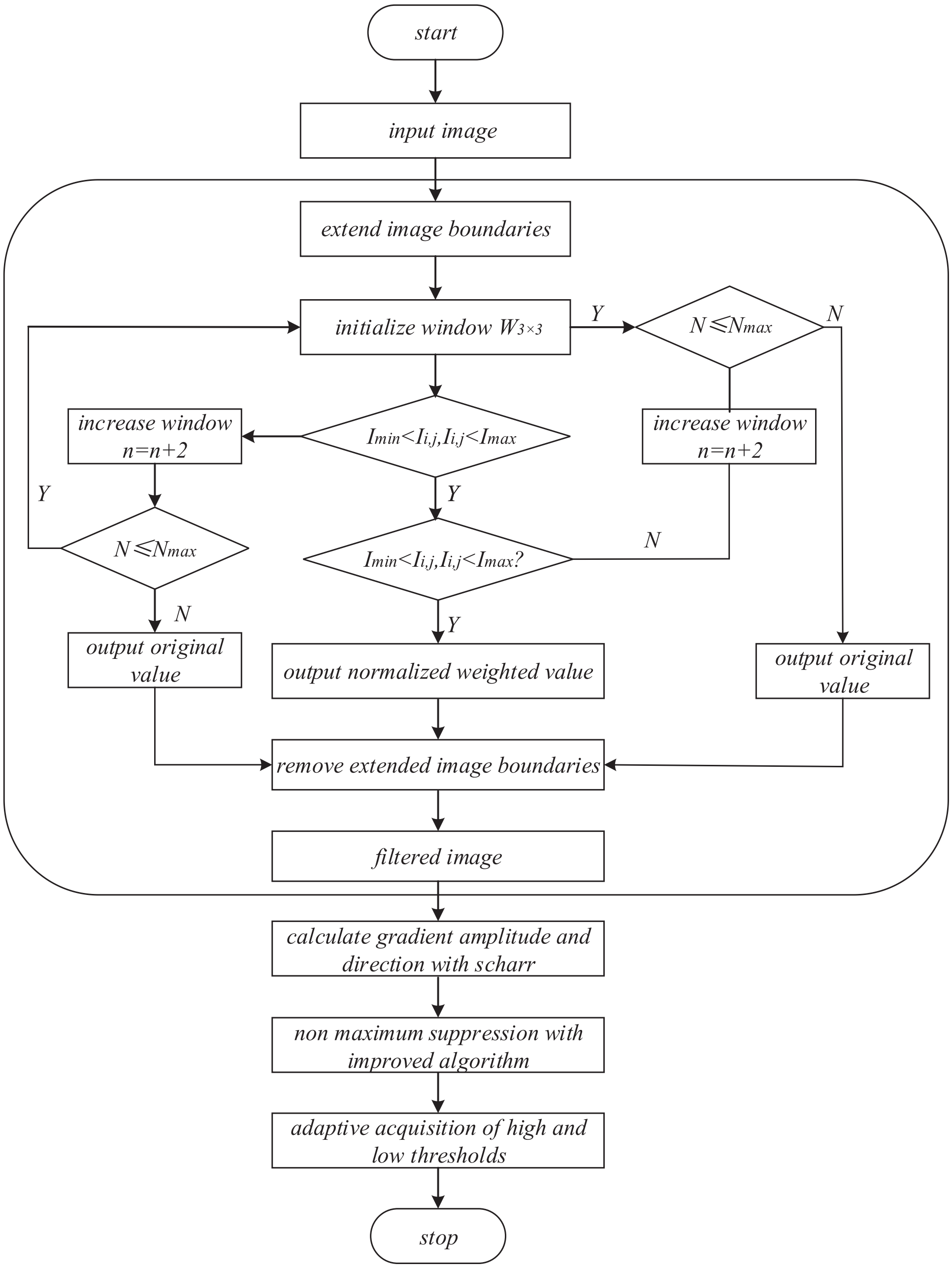

FIGURE 7 Flow chart of improved operator of Canny.

where N_r is the total number of valid elements in the window, I_{mean} is the mean value of the effective elements in the window.

3.2 | Gradient magnitude and direction

Traditional Canny edge detection uses the Sobel operator, which is slightly insufficient for weak edge detection. Therefore, the Scharr operator is introduced to enhance the gap between pixel values and enhance edge detection. Adjust the weight coefficient of the Sobel operator to obtain the Scharr operator, which is used to detect the magnitude and direction of the x-axis and y-axis gradient of the image, and achieve lateral and longitudinal edge value estimation. The magnitude and direction of the gradient are shown in Equations (18) and (19).

$$S_x = \begin{bmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{bmatrix} \tag{18}$$

$$S_y = \begin{bmatrix} -3 & -10 & -3 \\ 0 & 0 & 0 \\ 3 & 10 & 3 \end{bmatrix} \tag{19}$$

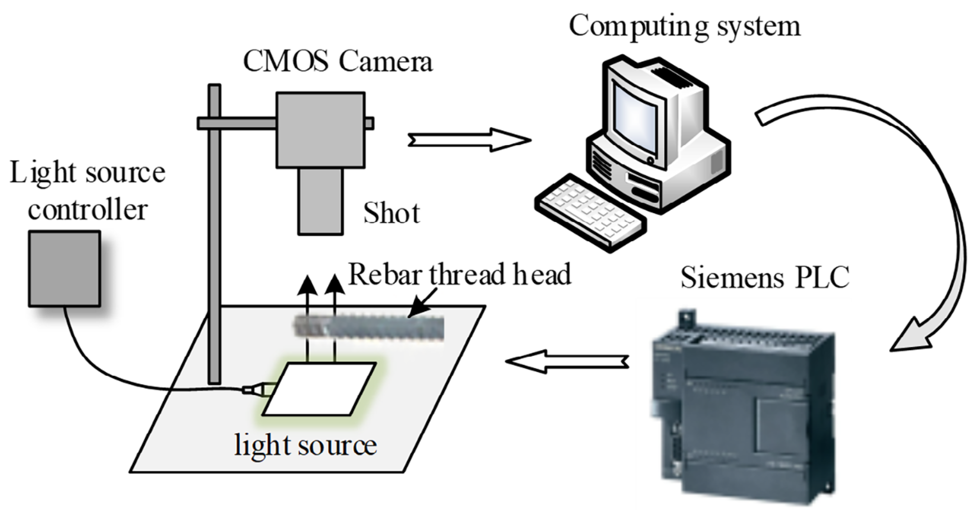

FIGURE 8 The platform configuration.

Substitute them into Equations (20) and (21) to find the gradient direction of each pixel in the image θ and amplitude G.

$$\theta = \arctan\left(\frac{S_y}{S_x}\right) \tag{20}$$

$$G = \sqrt{S_x^2 + S_y^2} \tag{21}$$

Due to the need to detect threaded edges, templates in 45° and 135° directions are added. The gradient template can be shown in Equations (22) and (23).

$$S_{45^\circ} = \begin{bmatrix} 10 & 3 & 0 \\ 3 & 0 & -3 \\ 0 & -3 & -10 \end{bmatrix} \tag{22}$$

$$S_{135^\circ} = \begin{bmatrix} 0 & -3 & -10 \\ 3 & 0 & -3 \\ 10 & 3 & 0 \end{bmatrix} \tag{23}$$

The evolved gradient amplitude formula can be shown in Equations (24).

$$G = |S_x| + |S_y| + |S_{45^\circ}| + |S_{135^\circ}| \tag{(24)}$$

where S_{45°, S_{135° are the gradient amplitudes after convolution.

3.3 | Improved non maximum suppression

An important step in the process of image edge detection is to locate image edge points and reduce the number of false edges. In the process of non-maximum suppression, it is necessary to obtain local maximum values through interpolation and comparison to obtain edge points. The non-maximum suppression interpolation step of the traditional Canny operator has great randomness in selecting interpolation points, resulting in inaccurate edge detection. In the process of non-maximum suppression, the traditional Canny operator is shown in Equation (25) during the interpolation process.

$$\begin{aligned} M_1 = {} & M_y \times I\,(i + adds\,(2)\,, j + adds\,(1)) + \left(M_x - M_y\right) \\ & \times I\,(i + adds\,(4)\,, j + adds\,(3)) \end{aligned} \tag{25}$$

where M_x is the coordinate of the Scharr horizontal gradient direction of the image, M_y is the coordinate of the Scharr vertical gradient direction of the image, I is the input image, and $adds()$ is the edge normal angle, M_1 is an interpolation point in the gradient direction, rotate pi, a symmetric interpolation point M_2 can be obtained with the same method.

Aiming at the arbitrary defect of selecting interpolation points in traditional algorithms, an improved interpolation method with non-maximum suppression is put forward. The formula is shown in Equation (26).

$$\begin{aligned} M_{11} = {} & M_y \times I\,(i + adds\,(2)\,, j + adds\,(1)) + \alpha(\left(M_x - M_y\right) \\ & \times I\,(i + adds\,(4)\,, j + adds\,(3)) \\ & - M_y \times I\,\left(i + adds\,(2)\,, j + adds\,(1)\right)) \end{aligned} \tag{26}$$

In Equation (26), an interpolation proportion coefficient α is added. The other coordinate point is not directly obtained through an interpolation function alone, but is calculated using the difference between the coordinate point of the interpolation function and the obtained interpolation point multiplied by α times. In this way, the error between two coordinate points will significantly reduce, which will make the interpolation point M_{11} in the gradient direction more accurate, edge points more refined, and position more accurate.

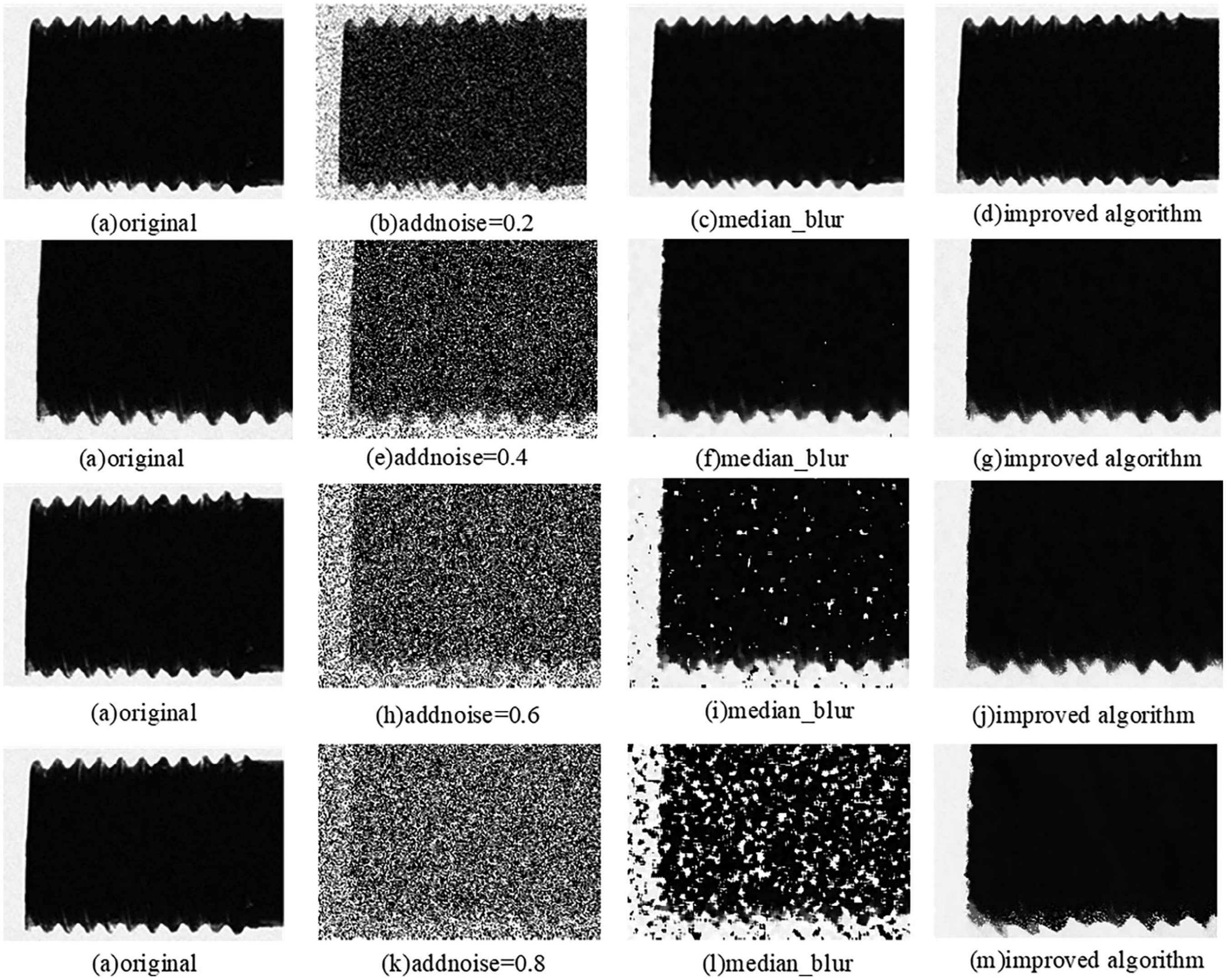

FIGURE 9 Comparison results of four groups of experiments on filtering rebar thread head image. (a) original; (b) addnoise=0.2; (c) median_blur; (d) improved algorithm; (e) addnoise=0.4; (f) median_blur; (g) improved algorithm; (h) addnoise=0.6; (i) median_blur; (j) improved algorithm; (k) addnoise=0.8; (l) median_blur; (m) improved algorithm.

TABLE 4 Filtering effect evaluation.

			addnoise = 0.2	addnoise = 0.4	addnoise = 0.6	addnoise = 0.8
Median-blur	MSE	Value	88.268	185.079	1425.708	9674.026
		Time (s)	0.213	0.431	0.267	0.211
	PSNR	Value	28.673	25.457	16.590	8.275
		Time (s)	0.001	0.001	0.001	0.001
	SSIM	Value	0.912	0.882	0.605	0.108
		Time (s)	0.010	0.017	0.013	0.011
Improved algorithm	MSE	Value	41.674	112.741	302.967	2247.355
		Time (s)	0.002	0.002	0.002	0.001
	PSNR	Value	31.932	27.610	23.317	14.614
		Time (s)	0.002	0.001	0.001	0.001
	SSIM	Value	0.971	0.943	0.892	0.743
		Time (s)	0.004	0.004	0.005	0.005

3.4 | Otsu confirms the high and low threshold

Otsu algorithm is a binary threshold segmentation algorithm [41]. According to the grayscale characteristics of the image, it sets the segmentation threshold to Th, and divides the pixels on the image into foreground $C1$ and background $C2$. The average greyscale of the foreground $C1$ is $m1$, and the ratio of these pixels to all pixels in the image is $p1$; The average grey level of the background $C2$ is $m2$, and the ratio of these pixels to all pixels in the image is $p2$. The global grayscale mean of the image is mG, and the inter class variance is recorded as σ. The relationships between them are shown in Equations (27) and (28).

$$p1 \times m1 + p2 \times m2 = mG \tag{27}$$

$$p1 + p2 = 1 \tag{28}$$

According to the concept of variance, the expression of inter class variance can be described in Equation (29).

$$\sigma = p1 \times (m1 - mG)^2 + p2 \times (m2 - mG)^2 \tag{29}$$

Optimize the expression, and the result is shown in Equation (30).

$$\sigma = p1 \times p2(m1 - m2)^2 \tag{30}$$

Calculate the intra class variance of all pixels from 0 to 255, and obtain the maximum value of Equation (30), which is the high threshold Th. Take the low threshold Tl as half of the high threshold Th. The Otsu method for solving the threshold can avoid the problem of poor adaptation caused by manually setting the threshold.

3.5 | Step of edge detection with improved Canny operator

An improved adaptive weighted median filter is used to remove noise, and then the gradient amplitude and direction of the image can be obtained through the Scharr operator. On this basis, the improved non-maximum value is used to suppress thinning edges, and finally, the adaptive extraction of high and low thresholds is achieved through Ostu to ultimately achieve image edge detection. The algorithm flowchart is shown in Figure 7.

4 | RESULTS AND DISCUSSION

4.1 | Experimental environment

To verify the advantages of the algorithm proposed here for edge detection of rebar thread head images, the platform configuration for this experiment is shown in Figure 8. The hardware

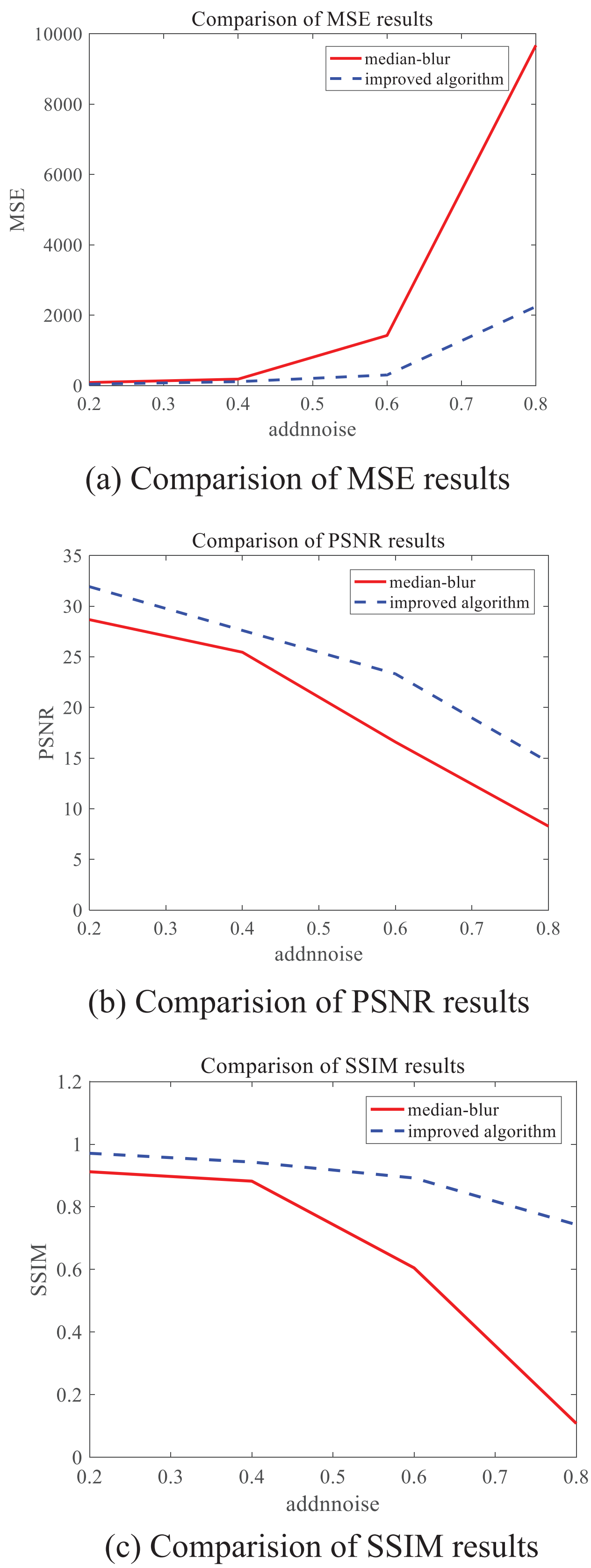

FIGURE 10 Comparison of image effect evaluation. (a) Comparision of MSE results; (b) Comparision of PSNR results; (c) Comparision of SSIM results.

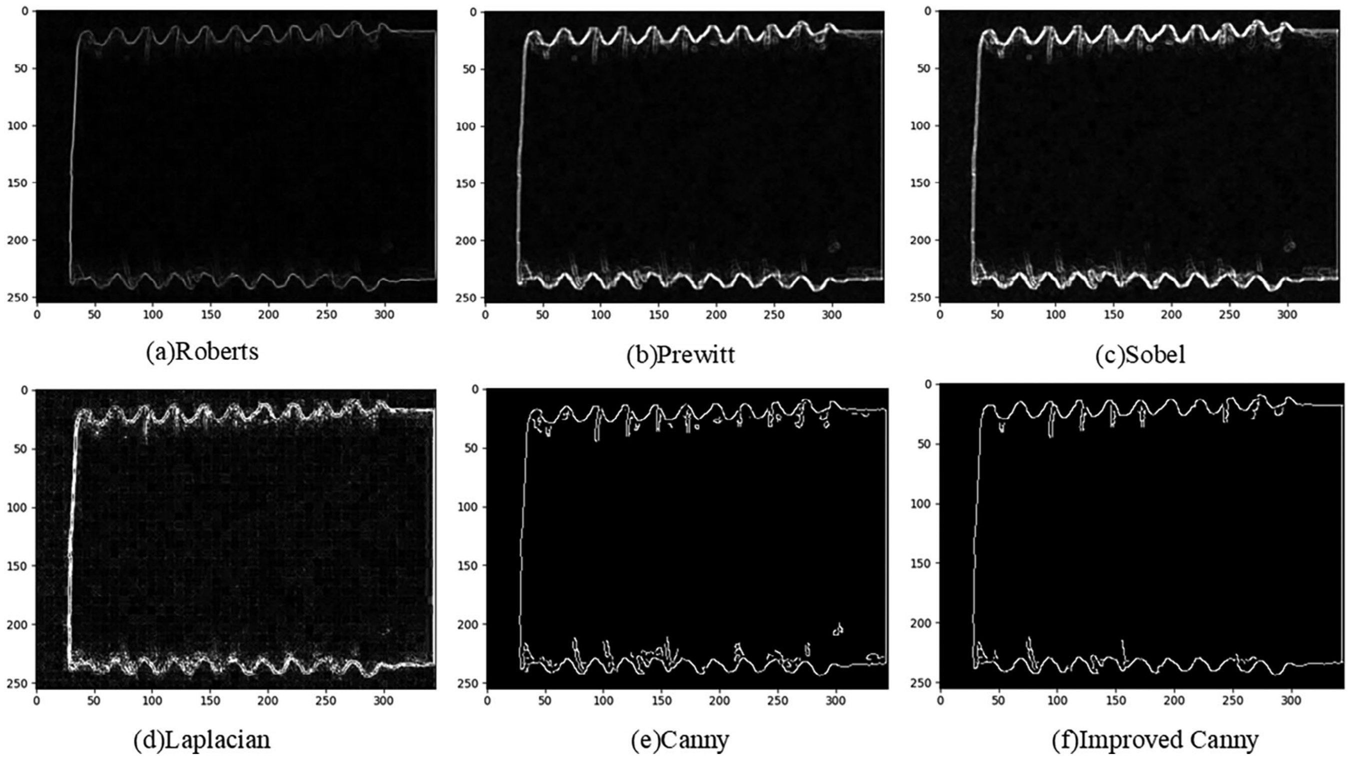

FIGURE 11 Comparison of edge detection effects of different operators. (a) Roberts; (b) Prewitt; (c) Sobel; (d) Laplacian; (e) Canny; (f) Improved Canny.

model is Intel (R) Core (TM) i5-8265U CPU @ 1.60 GHz with memory of 1.80 GHz and built-in 8.00 GB RAM. The running system is Windows11, and running software is pycharm, implemented based on python 3.7.4 and OpenCV4.7.0.68. The external equipment mainly includes a Haikang industrial camera whose model is MV-CE100-30GM and a square backlight light source.

4.2 | Experimental results

4.2.1 | Filtering and smoothing test for rebar thread head images

An improved adaptive median filter was used instead of a Gaussian filter to smooth the rebar thread head image, with a minimum window size of 3 and a maximum window size of 9. The experimental comparison results of median filtering and the algorithm proposed here with the noise 20%, 40%, 60%, and 80%, respectively were shown Figure 9. MSE, PSNR, and SSIM were used to evaluate the filtering effect. The data was shown in Table 4, and the comparison diagram was shown in Figure 10.

As can be seen from Figure 9, with the increase of noise concentration, traditional median filtering edges would produce blurring, and at the same time, more and more noise remained in the background, which could not be well suppressed and removed. The improved algorithm proposed here preserved edge information while filtering, and effectively suppressed noise. The larger the noise concentration, the more obvious the contrast effect was. From Table 4 and Figure 10, it can be seen that compared to the effect of median filtering, the algorithm proposed here had a smaller MSE, a larger PSNR, and a larger SSIM after filtering the image, indicating that the image processed by the improved algorithm was closer to the original image and has a better effect. Moreover, the greater the noise concentration, the greater the difference in results was, and the more obvious the contrast effect was. In addition, in terms of time, compared to the median filtering speed, the improved algorithm was also faster.

4.2.2 | Experiments on edge detection of rebar thread head image

Qualitative analysis of edge detection

In order to verify the advantages of the edge detection algorithm proposed here, use Figure 9a as the original image, the edge detection experiment was performed. Compare it with those processed by common edge detection operators, the results were shown in Figure 11. As can be seen from the figure, the edge details of the image processed by Robert's algorithm were severely lost and the extraction accuracy was not high. Although the edges detected by Prewitt and Sobel algorithms were relatively smooth in the overall image and had a clear outline, however, some of the detailed edges in the original image had not been well preserved after processing. The Laplacian algorithm could detect relatively rich edges, but it was not sensitive to noise and the detection effect was relatively fuzzy. The

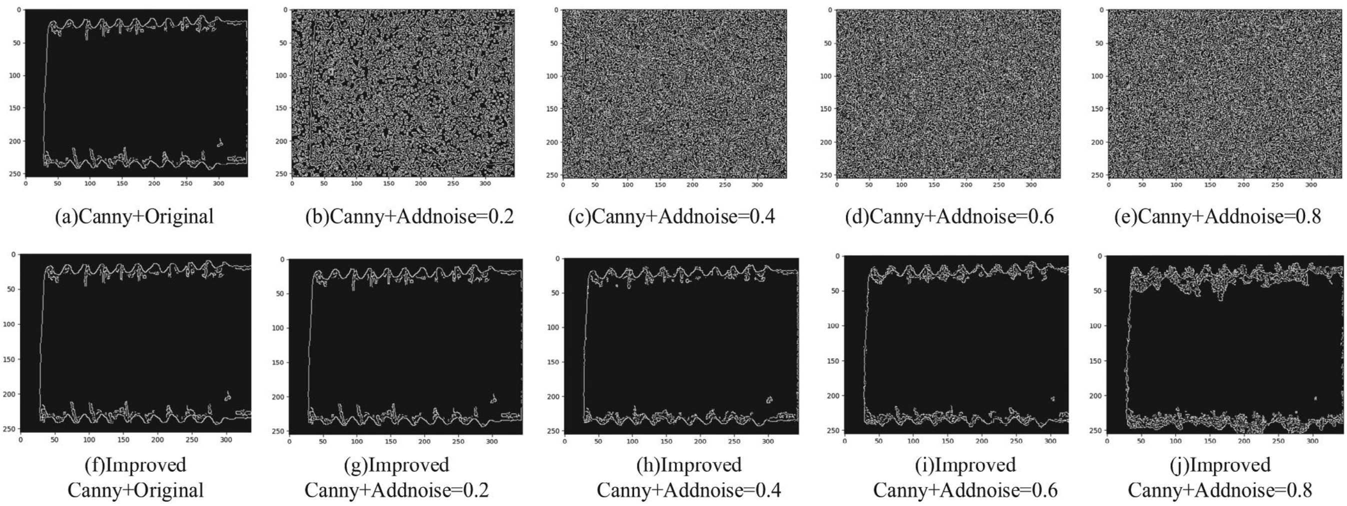

FIGURE 12 Comparison of edge detection effects on rebar thread head images added with different intensities of salt and pepper noise. (a) Canny+Original; (b) Canny+Addnoise=0.2; (c) Canny+Addnoise=0.4; (d) Canny+Addnoise=0.6; (e) Canny+Addnoise=0.8; (f) Improved Canny+Original; (g) Improved Canny+Addnoise=0.2; (h) Improved Canny+Addnoise=0.4; (i) Improved Canny+Addnoise=0.6; (j) Improved Canny+Addnoise=0.8.

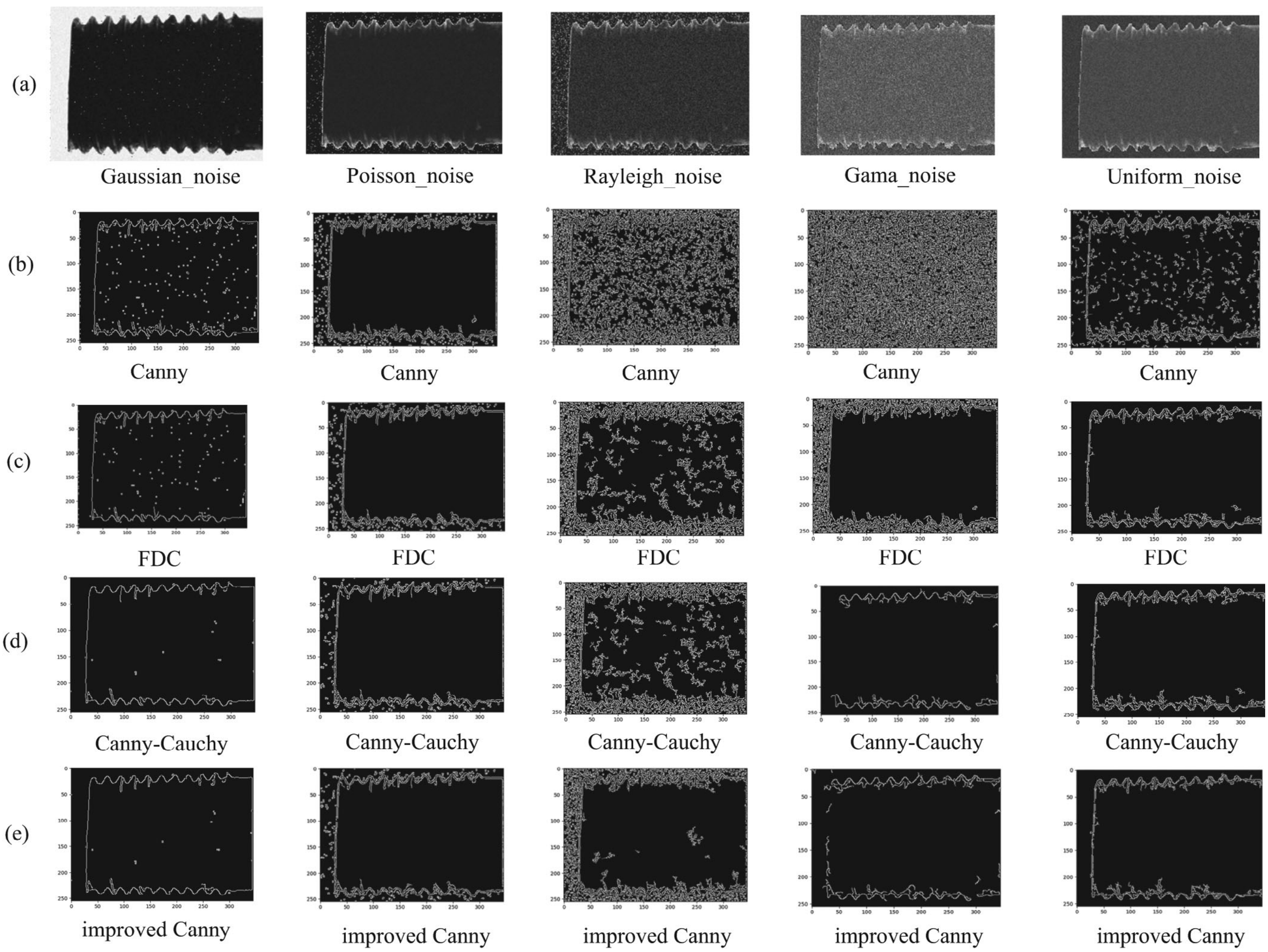

FIGURE 13 Comparison of edge detection effects on rebar thread head images added with different types of noise. (a) The results of adding different noises to the original image; (b) The results processed by Canny algorithm; (c) The results processed by FDC algorithm from other literatures; (d) The results processed by Canny-Cauchy algorithm from other literatures; (e) The results processed by improved Canny in this paper.

TABLE 5 Quantitative comparison of edge detection effects of two algorithms on different intensities of noisy images.

			addnoise = 0.2	addnoise = 0.4	addnoise = 0.6	addnoise = 0.8
Canny	AP	Value	0.364	0.301	0.276	0.187
		Time (s)	0.32	0.317	0.261	0.213
	ODS	Value	0.501	0.374	0.331	0.308
		Time (s)	0.001	0.001	0.001	0.001
	OIS	Value	0.534	0.112	0.104	0.103
		Time (s)	0.012	0.012	0.012	0.012
Improved Canny	AP	Value	0.921	0.838	0.773	0.751
		Time (s)	0.318	0.303	0.233	0.116
	ODS	Value	0.602	0.534	0.521	0.519
		Time (s)	0.001	0.001	0.001	0.001
	OIS	Value	0.610	0.656	0.635	0.627
		Time (s)	0.011	0.011	0.011	0.01

traditional Canny algorithm and the improved algorithm were superior to the other four algorithms. The former could extract relatively complete image edges, but it was sensitive to noise, resulting in some continuous noise contours, and generating more false edges. Compared to it, the improved algorithm was insensitive to noise, with continuous lines and no clutter, which could ensure the integrity of the edges, detect clear edges, and have better image detection results.

To verify the anti-noise performance of the improved algorithm, the traditional Canny algorithm and the improved algorithm were used to extract the edge of the original rebar thread head image, and added pepper and salt noise with the concentration of 0.2, 0.4, 0.6, and 0.8 to the original image in turn. Use the above two algorithms to detect the edge of the noisy image, as shown in Figure 12. Where, Figures 12a–e are the results after processing by the traditional Canny algorithm, and Figures 12f–j are the results processed by the improved Canny. As can be seen from the figure, the traditional Canny was greatly affected by noise, making the image edges blurred, so that almost no edge information could be detected. As the image noise concentration increased, the improved algorithm presented some noise contours, but it was less affected by noise and had a good suppression effect on noise, preserving the integrity and clarity of the edges, indicating that the algorithm proposed here had better noise resistance and robustness.

In order to further verify the anti-noise performance and robustness of the improved algorithm, traditional Canny, improved algorithm proposed here, and other algorithms in literatures were used to extract the edges of the original rebar thread head captured under backlight, and different types of noise were sequentially added to the original image, such as Gaussian noise, Poisson noise, Rayleigh noise, Gamma noise, Uniform noise etc. Use the above algorithms to detect the edges of noisy images, as shown in Figure 13. Here, Figure 13a is the result of adding different noises to the original image, Figure 13b is the result of traditional Canny algorithm, Figure 13c [12] and Figure 13d [42] are the results of edge extraction from other literatures, and Figure 13e is the processing result of the improved Canny algorithm proposed here. From Figure 13, it could be seen that traditional Canny was greatly affected by noise, and the processing results for different types of noise were not ideal, resulting in blurred image edges. The algorithm in Figure 13c was better at obtaining image edges than the Canny algorithm, but due to the limited number of image samples and noise, the effect of edge extraction was also unsatisfactory. The algorithm in Figure 13d had significantly better performance than Figure 13b,c, but for gamma noise, the end edge of the thread head image was not detected. Compared with other algorithms, the improved algorithm proposed here presented some noise contours, but had a better suppression effect on noise and was less affected by noise. For edge extraction of images with Gamma noise added, although the end face also showed edge breakage, the effect was significantly better than Figure 13d, indicating that the algorithm proposed here ensured noise reduction effect while maximizing the protection of image structure from being changed.

Quantitative comparison of experimental results

The evaluation of edge detection has been clearly defined in previous work [12]. Here, the evaluation indicators mainly include optimal dataset size (ODS), optimal image size (OIS), and average accuracy (AP). Among them, ODS mainly evaluates the performance threshold level of edge detection algorithms in different situations. The higher the score, the more correctly detected edge pixels; OIS is a variant of ODS aimed at finding the optimal threshold for maximizing edge detection. The larger the value, the stronger the significance of the edges, and the better the threshold selection; AP mainly evaluates the balance between accuracy and recall under different thresholds, and the larger its value, the greater the probability of detecting true boundary pixels in edge detection. It should be noted that unnecessary background information in the image can affect the above three evaluation indicators to a certain extent.

In order to further confirm the effectiveness of the algorithm proposed here, ODS, OIS, AP, and time were used to evalu-

TABLE 6 Quantitative comparison of edge detection effects of different algorithms on different types of noisy images.

		Gaussian	Poisson	Rayleigh	Gama	Uniform
Canny	AP	0.536	0.487	0.495	0.491	0.531
	ODS	0.519	0.421	0.173	0.219	0.431
	OIS	0.521	0.458	0.219	0.207	0.446
FDC	AP	0.581	0.562	0.599	0.643	0.641
	ODS	0.561	0.549	0.587	0.572	0.621
	OIS	0.532	0.539	0.541	0.568	0.665
Canny-Cauchy	AP	0.656	0.553	0.598	0.479	0.603
	ODS	0.639	0.567	0.584	0.492	0.621
	OIS	0.639	0.558	0.572	0.501	0.631
Improved Canny	AP	0.671	0.628	0.607	0.542	0.682
	ODS	0.657	0.601	0.563	0.526	0.661
	OIS	0.668	0.643	0.637	0.544	0.701

ate the edge detection effect compared the traditional Canny algorithm with the different algorithms. Firstly, process the images with different intensities of salt and pepper noise in Figure 12, the results were shown in Table 5. As can be seen from Table 5, compared to other edge detection algorithms, the improved algorithm had a larger AP, a larger ODS and a lager OIS, which indicated that the edge detection method proposed in this article was less susceptible to noise and had a stronger response to true edges. In addition, in terms of time, the improved algorithm was also the faster. Secondly, different algorithms were used to process the images with different types of noise in Figure 13, and the quantitative results obtained were shown in Table 6. From the results in Table 6, it can be seen that compared to other algorithms, the algorithm proposed here performed better in edge extraction for different types of noise, with higher accuracy in edge pixel recognition, and better continuity of the edges obtained.

5 | CONCLUSION

Due to the influence of external and other objective factors during the processing of rebar thread head, the image noise is large, which is not conducive to the edge feature extraction, and seriously affects the subsequent parameter acquisition and quality detection of thread head. Therefore, this paper further studies the Canny algorithm, directed to the problems of sensitivity to noise, existence of false edges, and lack of self-adaptability of the traditional Canny algorithm. Some of suggestions are proposed to improve the effects and efficiency of defect detection, which are summarized as below:

1. By comparing different filtering algorithms, an improved adaptive median filter was selected to replace Gaussian filter for image noise reduction.
2. Considering the shortcomings of Sobel operator in weak edge detection, Scharr operator is introduced to replace Sobel operator to enhance the gap between pixel values and enhance edge detection. At the same time, 45° and 135° directional templates are added to prevent edge information loss.
3. Aiming at the randomness of the selection of interpolation points in traditional algorithms, a scale factor is introduced to make the improved interpolation method with non-maximum suppression more accurate, the edge points are refined, and the positioning is more accurate.
4. Finally, Ostu realizes adaptive extraction of high and low threshold values, and finally realizes adaptive edge detection of the image.

Through simulation experiments for comparison and analysis, the effectiveness of the improved algorithm has been verified from both qualitative analysis and quantitative evaluation. The results show that the improved algorithm not only has good anti-noise performance in visual effects, but also obtains clear and smooth contour of the edge image after detection, with prominent detailed edges, and is superior to other algorithms in speed. All these facts indicate that the improved algorithm can achieve real-time adaptive edge extraction of rebar thread head in noisy environments without sacrificing edge detection accuracy, and can be applied to online quality detection. Future research can focus on how to extract the size parameters of the thread head to further complete quality detection based on the implementation of edge detection.

AUTHOR CONTRIBUTIONS

Li Liu: Conceptualization; funding acquisition; methodology; writing—original draft; writing—review and editing. **Zijin Liu**: Project administration; supervision; funding acquisition; writing—review and editing. **Aishan Hou**: Project administration; resources. **Xuefei Qian**: Data curation; formal analysis; writing—original draft; writing—review and editing. **Hui Wang**: Data curation; software; validation.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data subject to third party restrictions. The data that support the findings of this study are available from CABR Construction Machinery Technology Co., Ltd. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the corresponding author with the permission of CABR Construction Machinery Technology Co., Ltd.

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